SANOQ SISTEMALARI

Ikkilik • Sakkizlik • O'nlik • O'n oltilik

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0b10 • 0o10 • 10 • 0x10

Dars Rejasi

1. Sanoq sistemalariga kirish — nima va nima uchun?
2. Ikkilik sistema (Binary) — 0 va 1 dunyosi
3. Sakkizlik sistema (Octal) — Unix/Linux tarixi
4. O'nlik sistema (Decimal) — kundalik hayot
5. O'n oltilik sistema (Hexadecimal) — kiberxavfsizlikda asosiy
6. Sistemalar orasida konvertatsiya usullari
7. Linux amaliyoti — bc, printf, Python, xxd
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Sanoq Sistemasi Nima?

Sanoq sistemasi — raqamlarni yozish va ifodalash usuli.

Asos (Base)

Sistemadagi raqamlar soni.
Base-2: faqat 0 va 1
Base-16: 0-9 va A-F

Pozitsion qiymat

Har bir raqamning o'rni uning qiymatini aniqlaydi.
1011 ≠ 1101

Kompyuter va ikkilik

Kompyuterlar faqat elektr signali biladi:
0 = o'chiq, 1 = yoniq

Kiberxavfsizlik

Hex dump, shellcode, MAC, IP, SHA hashlar — barchasi shu asosda

O'nlik Sistema (Decimal)

Base-10 | Kundalik hayotimiz

O'nlik Sistema — Base 10

Raqamlar: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 — oddiy hisoblash tizimimiz

2025_{10} ni tushunish:

10^3	10^2	10^1	10^0
2	0	2	5
2000	0	20	5

$2000 + 0 + 20 + 5 = 2025_{10}$ ✓

O'nlikdan Ikkilikka — bo'lish usuli:

13 ÷ 2 = 6 qoldiq 1 ↑
6 ÷ 2 = 3 qoldiq 0 |
3 ÷ 2 = 1 qoldiq 1 |
1 ÷ 2 = 0 qoldiq 1 |
Qoldiqlarni teskari o'qish:
$13_{10} = 1101_2$

25 ÷ 2 = 12 q=1
12 ÷ 2 = 6 q=0
6 ÷ 2 = 3 q=0
3 ÷ 2 = 1 q=1
1 ÷ 2 = 0 q=1
$25_{10} = 11001_2$

Qoida: bo'lib, qoldiqni yig' va teskari o'qi

Ikkilik Sistema (Binary)

Base-2 | Faqat 0 va 1 | Kompyuterning tili

Ikkilik Sistema — Base 2

Faqat ikkita raqam: 0 va 1 | Har bir raqam = 1 bit

1011_2 = ? Pozitsion qiymatlar:

$2^3=8$	$2^2=4$	$2^1=2$	$2^0=1$
1	0	1	1
1×8=8	0×4=0	1×2=2	1×1=1

Jami: 8 + 0 + 2 + 1 = 11_{10}

Tezkor qo'llanma:

0000 = 0	1000 = 8
0001 = 1	1001 = 9
0010 = 2	1010 = 10
0011 = 3	1011 = 11

0100 = 4	1100 = 12
0101 = 5	1101 = 13
0110 = 6	1110 = 14
0111 = 7	1111 = 15

Bit = 1 raqam
Nibble = 4 bit
Byte = 8 bit
65536 = 2^{16} = 0xFFFF+1

Sakkizlik Sistema (Octal)

Base-8 | 0 dan 7 gacha | Linux chmod

Sakkizlik Sistema — Base 8

Raqamlar: 0, 1, 2, 3, 4, 5, 6, 7 (8 va 9 mavjud emas!)

52_8 = ? Pozitsion hisob:

$8^1=8$	$8^0=1$
5	2
5×8=40	2×1=2

Jami: 40 + 2 = 42_{10}

Linux chmod — sakkizlikning amaliy qo'llanmasi:

```
chmod 755 skript.sh
# 7 = rwx (egasi: o'qi+yoz+ijro)
# 5 = r-x (guruh: o'qi+ijro)
# 5 = r-x (boshqalar: o'qi+ijro)
```

r = 4 (0b100)
w = 2 (0b010)
x = 1 (0b001)

rwx = 4+2+1 = 7
rw- = 4+2+0 = 6
r-x = 4+0+1 = 5
r-- = 4+0+0 = 4

O'n Oltilik Sistema (Hexadecimal)

Base-16 | Kiberxavfsizlikda asosiy vosita

O'n Oltilik Sistema — Base 16

Raqamlar: 0–9, A, B, C, D, E, F (A=10, B=11, C=12, D=13, E=14, F=15)

Hex:	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Dec:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Misol: $0x1F_{16}$ = ?

$16^1=16$	$16^0=1$
1	F(=15)
$1×16=16$	$15×1=15$

Jami: 16 + 15 = 31_{10}

0xDEAD = ?
D=13, E=14, A=10, D=13
$= 13×16^3 + 14×16^2 + 10×16 + 13$
$= 53248+3584+160+13 = 57005_{10}$

Kiberxavfsizlikda: 0xDEAD, 0xBEEF, 0xCAFE, 0xFEED — mashxur magic bytes!

To'liq Konvertatsiya Jadvali

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Konvertatsiya Usullari

O'nlik → Ikkilik:
13 ÷ 2 = 6 q=1
6 ÷ 2 = 3 q=0
3 ÷ 2 = 1 q=1
1 ÷ 2 = 0 q=1
→ 1101_2

Ikkilik → Hex:
11011010_2
→ 1101 | 1010
→ D | A
→ 0xDA

Hex → Ikkilik:
0xFF
F=1111 F=1111
→ 11111111_2
= 255_{10}

Tezkor usul: Har 1 hex raqam = 4 bit

0=0000 1=0001 2=0010 3=0011
4=0100 5=0101 6=0110 7=0111
8=1000 9=1001 A=1010 B=1011
C=1100 D=1101 E=1110 F=1111

Misol: 0xCAFE = ?
C=1100 A=1010 F=1111 E=1110
→ $1100\ 1010\ 1111\ 1110_2$
= 51966_{10}

O'nlik → Hex (bo'lish usuli):

255 ÷ 16 = 15 q=F(15)
15 ÷ 16 = 0 q=F(15)
Teskari: FF_{16} = 255_{10}

Demak: 0xFF = 255
256 = 0x100
65535 = 0xFFFF

Linux Amaliyoti

bc, printf, Python, xxd — terminalda ishlash

Linux: bc — Base Calculator

bc — terminal kalkulyatori, ixtiyoriy asosda hisob-kitob imkoniyati

```
# bc o'rnatish
sudo apt install bc

# Ikkilikdan o'nlikka
echo 'ibase=2; 1101' | bc
# Natija: 13

# O'nlikdan ikkilikka
echo 'obase=2; 13' | bc
# Natija: 1101
```

```
# O'nlikdan hex ga
echo 'obase=16; 255' | bc
# Natija: FF

# Hex dan o'nlikka
echo 'ibase=16; FF' | bc
# Natija: 255

# Sakkizlikdan o'nlikka
echo 'ibase=8; 77' | bc
# Natija: 63
```

```
# Ko'p hisob-kitob (here-doc)
ibase=2
1010
1111
11001100
```

```
Hex dan ikkilikka - faylga yozganda
ibase=16
obase=2
FF
DEAD
```

Linux: xxd va hexdump — Fayl Tahlili

Fayl tarkibini hex shaklida ko'rish — kiberxavfsizlikda asosiy ko'nikma!

```
echo 'Salom' | xxd
# 00000000: 5361 6c6f 6d0a  Salom.

# Fayl hex dump
xxd /etc/hostname

# Faqat hex (ASCII yo'q)
xxd -p fayl.bin
# Binary faylga qayta yozish
echo '48656c6c6f' | xxd -r -p
```

```
# hexdump
hexdump -C fayl.bin

# od — turli formatlarda
od -c fayl.txt  # belgilar
od -x fayl.txt  # hex
od -b fayl.txt  # oktal

# strings — binary fayldan
# o'qiluvchi matnlarni topish
strings /bin/ls | head -20
```

Magic Bytes — Fayl Turini Aniqlash:

```
xxd /bin/ls | head -2 -> ELF (Linux bajariladigan fayl)
xxd rasm.png | head -1 ->  PNG rasm
xxd hujjat.pdf | head -1 ->  PDF hujjat
```

Fayl turi:
PDF: 25 50 44 46
PNG: 89 50 4E 47
JPG: FF D8 FF
ELF: 7F 45 4C 46
ZIP: 50 4B 03 04
GIF: 47 49 46 38

Kiberxavfsizlikda Qo'llanilishi

MAC, IP, Hash, XOR, Port, Shellcode

Kiberxavfsizlikda Sanoq Sistemalari

MAC Manzillar

00:1A:2B:3C:4D:5E
→ Har juftlik = 1 byte (hex)
→ Qurilmani aniqlash

IP Manzillar

192.168.1.1
→ Har qism = 0–255 (1 byte)
→ C0.A8.01.01 (hex)

SHA/MD5 Hashlar

MD5: 32 hex belgi = 128 bit
SHA256: 64 hex = 256 bit
Parol tekshirish

XOR Shifrlash

0x41 XOR 0x0F = 0x4E
A XOR ? = N
Oddiy shifrlash asosi

Port Raqamlari

0x0016 = 22 (SSH)
0x0050 = 80 (HTTP)
0x01BB = 443 (HTTPS)

Shellcode

\x90\x90\x31\xc0
NOP NOP XOR EAX
Assembly→Hex→Payload

Amaliy Mashqlar — Terminalda Bajaring!

Quyidagi 6 ta mashqni terminalda bajarib ko'ring:

Mashq 1: bc bilan

```
echo 'obase=2; 42' | bc
echo 'ibase=16; DEAD' | bc
echo 'obase=16; 255' | bc
```

Mashq 2: Python3

```
python3 -c "print(hex(255),bin(255))"
python3 -c "print(int('FF',16))"
python3 -c "print(0xDEAD)"
```

Mashq 3: printf

```
printf '%x\n' 255
printf '%b\n' 10
printf '%o\n' 8
```

Mashq 4: xxd

```
echo 'CTF' | xxd
xxd /etc/hostname
printf '\xDE\xAD\xBE\xEF' | xxd
```

Mashq 6: IP manzil

```
printf '%d.%d.%d.%d\n' 0xC0 0xA8 0x01 0x01
# 192.168.1.1
```

Xulosa va Takrorlash

Ikkilik Base-2	Sakkizlik Base-8	O'nlik Base-10	Hex Base-16
Faqat 0 va 1 Kompyuter tili 8 bit = 1 byte	0 dan 7 gacha Linux chmod 3 bitli guruh	Kundalik hayot 0 dan 9 gacha Insonga tabiiy	0–9, A–F Kiberxavfsizlik 4 bitli nibble